ChatGPT

Website Review: Aarit Malhotra's Portfolio

Aarit Malhotra's personal portfolio (hosted on Vercel) showcases his work and contact details through a single-page application with a dark, animated aesthetic. The site displays a pre-loader, a hero section, an *About* section, a list of projects, and a *Get in Touch* section. This review covers user interface (UI/UX), accessibility, performance and search-engine optimisation (SEO). Evidence is drawn from Google PageSpeed Insights and recognised standards from Google Search Central, WCAG 2.1 and modern UI design guidelines.

1 UI/UX Evaluation

1.1 Visual design

- **Aesthetic and palette** – The site uses a dark radial-gradient background with animated particles. The hero heading ("Hey! I'm Aarit") is large and striking. Although visually appealing, the high-contrast gradients draw attention away from content and can reduce readability on low-brightness screens. Modern design guides recommend balancing visual interest with *white-space* and intentional grouping; micro-whitespace improves readability between words and paragraphs and macro-whitespace around headings makes layouts feel less cluttered [1]. Currently the dark background and particle effects leave little white-space.

- **Consistent spacing and grouping** – The portfolio items are separated by thin horizontal lines and numbers. However, there is little vertical spacing between sections, causing content to feel cramped. According to web-design best practices, space should be used to group related elements and separate unrelated ones [2]. Adding padding between sections (e.g., between *About*, *Portfolio* and *Get in Touch*) would improve readability.

- **Navigation** – A floating navigation bar appears near the top of the page. When scrolling, the menu sometimes overlaps the content (observed in the portfolio section), which can be distracting. A clearer sticky header with appropriate z-index and background transparency would avoid overlap and keep navigation accessible. UX best-practice guidance suggests limiting primary menu items to 5–8 and using descriptive labels [3]; the current menu follows this but could benefit from larger click targets and a clear visual indication of the current section.

- **Hero section and pre-loader** – The pre-loader displays a glowing logo and a progress bar before the main page appears. PageSpeed Insights reported a poor Largest Contentful Paint (LCP) of **5.7 s** and a Speed Index of **6.8 s** [4], suggesting that the pre-loader and heavy animations delay the initial render. Removing or shortening the pre-loader and deferring non-essential scripts would improve perceived performance. Using a static hero image (as recommended for engaging landing pages [5]) and lazy-loading non-critical assets would also help.

- **Projects section** – Each project is listed by name with a short description ("Development / Design", "UI/UX Design / Fullstack Development", etc.). However, the project names are not hyperlinks and there is no call-to-action. For a portfolio, users expect to click through to individual project pages or modals showing details, technologies used and outcomes. Adding interactive cards with descriptive alt text and links to project pages would make the portfolio more engaging.

- **Contact section** – Contact information is split between two columns: email/phone/location on the left and social-media links on the right. To improve usability on mobile devices, ensure the phone number uses a `tel:` link and the email address uses a `mailto:` link. A simple contact form (with proper labels) could allow visitors to send a message without switching apps.

1.2 Responsiveness and accessibility of UI

- **Responsiveness** – The site appears to be responsive, but the small floating menu may be difficult to tap on mobile devices. For small screens, consider a hamburger menu with larger touch targets (as suggested for mobile navigation [3]) and ensure sections stack vertically with sufficient spacing.

- **Keyboard navigation** – The PageSpeed Insights audit flagged that “buttons do not have an accessible name” and “links do not have a discernible name” [6]. Social-media icons and project entries should include visible text or `aria-label` attributes so that screen-reader and keyboard users understand their purpose. Focus styles should be clearly visible when navigating via keyboard.

2 Accessibility Review

Web accessibility is governed by the Web Content Accessibility Guidelines (WCAG 2.1). Key requirements include colour contrast, alternative text for images and descriptive names for controls.

2.1 Colour contrast

The WCAG *Contrast (Minimum)* success criterion requires a **contrast ratio of at least 4.5 : 1 for normal text** and **3 : 1 for large text** [7]. The portfolio uses pale text on a dark, colourful gradient. In some places (e.g., grey subheadings on blue backgrounds) the contrast may be below 4.5 : 1. To improve:

1. Use a colour-contrast checker to verify each text/background combination. Adjust colours or increase font weight where contrast fails, ensuring normal text meets 4.5 : 1 and headings meet 3 : 1 as required by WCAG [7].
2. Provide a light-mode option or allow users to toggle between dark and light themes.

2.2 Alternative text for images

The site uses icons and images in the pre-loader and contact section. The W3C tutorial on images states that all non-text content must have a text alternative describing its information or function [8]. Guidelines include:

- Provide a short description for **informative images** [9].
- Use `alt=""` for **decorative images** that do not convey information [10].
- For **functional images** (e.g., social-media icons used as links), describe their function rather than their appearance [11].

- Avoid embedding text within images; if images of text are used, the alt text should contain the same words [12].

Social icons (GitHub, LinkedIn, Instagram) currently lack accessible names [6]. Add `aria-label` attributes or visually hidden text so that screen-reader users know where the link leads. For decorative particle backgrounds, set `aria-hidden="true"` and `alt=""` so they are ignored by assistive technology.

2.3 Semantic markup

Use HTML5 semantic elements (`<header>`, `<nav>`, `<main>`, `<section>`, `<footer>`) and headings (`<h1>`..`<h3>`) to structure the page logically. A proper heading hierarchy aids screen readers and improves SEO. Ensure lists (e.g., the portfolio numbers) use ordered lists (`<ol>` / `<li>`) instead of plain divs to convey order semantically.

2.4 Keyboard and focus

Ensure every interactive element (nav links, icons, projects) is keyboard-reachable and displays a visible focus indicator. Add `tabindex="0"` where necessary and avoid trapping focus inside custom components. Provide skip-to-content links for quick navigation.

3 Performance Analysis

Google PageSpeed Insights (mobile test) rated the site with **Performance 68/100**, **Accessibility 90/100**, **Best Practices 96/100** and **SEO 100/100** [4]. Key performance metrics were:

Metric	Observed value	Target (Core Web Vitals)	Observation
Largest Contentful Paint (LCP)	**5.7 s** [4]	$\leq$ 2.5 s [13]	LCP is significantly above the recommended threshold, likely due to heavy animations and large hero content.
First Contentful Paint (FCP)	**1.8 s** [4]	Aim for < 1.8 s	On the threshold; could be improved by optimising CSS/JS delivery.
Total Blocking Time (TBT)	**200 ms** [4]	$\leq$ 200 ms	At the upper limit; reducing JavaScript execution time would help.
Speed Index	**6.8 s** [4]	Lower is better	Indicates users see content slowly; heavy animations and scripts contribute.
Cumulative Layout Shift (CLS)	**0.016** [4]	$\leq$ 0.1 [13]	Good; minimal unexpected layout shifts.

3.1 Performance bottlenecks

PageSpeed Insights diagnostics flagged several issues [14] :

1. **Excessive JavaScript execution and main-thread work** – JavaScript execution time is ~1.4 s and main-thread work ~2.8 s [14] . Removing unused code, splitting bundles and using dynamic imports can reduce blocking. Tools like `webpack` or `Next.js` 's `dynamic()` can lazy-load modules.
2. **Unused JavaScript** – ~136 KiB of unused JavaScript was detected [14] . Audit dependencies and remove unused libraries. Use tree-shaking and `next/font` for fonts.
3. **Long tasks and non-composited animations** – Six long main-thread tasks and 11 non-composited animations were found [14] . Replace heavy JavaScript animations (e.g., particle effects) with CSS or SVG animations, or limit them to the hero section; avoid animations that trigger layout or paint (use `transform` and `opacity`).
4. **Render-blocking resources** – There are render-blocking requests (CSS/JS) that delay FCP [14] . Use `<link rel="preload">` for critical resources, defer non-critical scripts and combine small CSS files. Ensure fonts are loaded efficiently (`font-display: swap`).
5. **Cache lifetimes & legacy JavaScript** – Some static assets lack caching and legacy JS bundles could be removed [14] . Configure caching headers via `next.config.js` / `vercel.json` and ensure modern builds.

3.2 Suggested performance improvements

- **Optimise images and media** – Compress images and export hero graphics in modern formats (e.g., WebP). Serve different sizes via `<picture>` or Next.js `Image` component with lazy loading. Replace the particle animation with a static gradient or reduce particle count.
- **Reduce pre-loader impact** – Skip the pre-loader or implement it using CSS only; ensure it does not block HTML rendering. Consider skeleton screens or progress indicators that do not delay content.
- **Minimise third-party scripts** – Only load analytics or external widgets after the first paint; use async/defer attributes.
- **Implement server-side rendering and caching** – Next.js can statically generate or server-render pages; this reduces initial JS load. Use incremental static regeneration for dynamic content.
- **Monitor Core Web Vitals** – Regularly test with PageSpeed Insights or Web Vitals and aim for LCP under 2.5 s, Interaction to Next Paint (INP) < 200 ms and CLS < 0.1 as recommended [13] .

4 Search-Engine Optimisation (SEO)

PageSpeed Insights awarded the site **SEO 100/100** [4] , but there are still areas to strengthen.

4.1 Meta tags and descriptive snippets

Google's guidelines state that a *meta description* should provide a short, relevant summary of a page; there is no strict length limit, but snippets are truncated to fit the device width [15] . Each page should have a unique description [16] and include relevant information about the content [17] . Ensure that:

- The `<title>` and `<meta name="description">` accurately describe the portfolio and include keywords like "Aarit Malhotra portfolio", "developer", etc.

- Meta descriptions avoid keyword stuffing and summarise the content clearly [18]. For example: `"Portfolio of Aarit Malhotra – high school student and developer. Projects, resume and contact information."`.
- Pages have unique `<title>` and `<meta>` tags. Even on a single-page app, unique titles/descriptions can be set for each section with dynamic routing or anchors.

4.2 Structured data and open-graph tags

Although the SEO audit passed, adding structured data improves search appearance. Use JSON-LD to mark up:

- **Person** or **Organization** schema with the developer's name, job title, image and social profiles.
- **Project** schema (e.g., `CreativeWork` or `SoftwareSourceCode`) for each portfolio item, including description, technologies and links.

Also include Open Graph (`og:title`, `og:description`, `og:image`, `og:type`) and Twitter Card tags so that sharing links on social media shows a preview.

4.3 Robots and sitemap

Ensure there is a `robots.txt` that allows search engines to crawl the site and an XML sitemap. Vercel or Next.js can generate a sitemap automatically. Submit the sitemap to Google Search Console.

4.4 Internal linking and anchors

The portfolio uses a single page with anchor links. To improve navigation and search indexing:

- Use descriptive anchor texts (e.g., `#about`, `#portfolio`) and update the URL as users scroll (using the History API). This allows users to share links to specific sections.
- Provide internal links to detailed project pages where possible. Rich content helps search engines understand the scope of each project.

5 Security and Best Practices

PageSpeed Insights flagged two general best-practice issues [19]:

- **Browser errors logged to console** – Fix console errors to avoid exposing internal information.
- **Missing source maps for large JS files** – Provide source maps during development but not in production to aid debugging; remove them from production builds if not needed.

Trust-and-safety guidelines suggest enabling a Content Security Policy (CSP) to mitigate cross-site scripting (XSS) and clickjacking. Configure CSP, COOP and other security headers in the deployment settings. Use HTTPS (provided by Vercel) for all resources.

6 Conclusion and Prioritised Recommendations

Aarit Malhotra's portfolio has a polished visual style and high SEO score, but it suffers from slow loading and some accessibility oversights. Implementing the following improvements will enhance usability, performance and search visibility:

1. **Optimise performance** – Remove or shorten the pre-loader, compress images, lazy-load non-critical assets, and reduce JavaScript bundle size and animations [4] [14]. Aim for LCP < 2.5 s and TBT < 200 ms [13].
2. **Improve colour contrast and whitespace** – Test and adjust text-to-background contrast to meet WCAG 4.5 : 1 and 3 : 1 ratios [7]. Increase vertical spacing between sections and group related items intentionally [1] [2].
3. **Add accessible names and alt text** – Ensure all buttons and links have descriptive labels and all images have appropriate alt text [8] [6]. Provide keyboard focus indicators and allow keyboard navigation.
4. **Enhance navigation and project details** – Make the navigation bar sticky without overlapping content, offer a mobile-friendly menu and link project names to detailed pages. Use descriptive anchor text for section links [3].
5. **Strengthen meta and structured data** – Write unique `<title>` and `<meta description>` tags per section, add Open Graph and Twitter Card tags and include JSON-LD structured data [20].
6. **Implement security headers** – Fix console errors, provide source maps only in development and set CSP and COOP headers [19].

Following these recommendations will create a more accessible, performant and user-friendly portfolio while retaining the distinctive visual identity.

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[1] [2] [5] Best Web Design Practices: UI/UX, Hierarchy, and More (2026)
https://www.hostinger.com/tutorials/web-design-best-practices

[3] Top UX Best Practices for 2025 Web Design - New Directions In Computing
https://www.ndic.com/blog/top-ux-best-practices-for-2025-web-design/

[4] [6] [14] [19] PageSpeed Insights
https://pagespeed.web.dev/analysis/https-aaritmalhotra-vercel-app/ag2ggghf4v

[7] Understanding Success Criterion 1.4.3: Contrast (Minimum) | WAI | W3C
https://www.w3.org/WAI/WCAG21/Understanding/contrast-minimum.html

[8] [9] [10] [11] [12] Images Tutorial | Web Accessibility Initiative (WAI) | W3C
https://www.w3.org/WAI/tutorials/images/

[13] Understanding Core Web Vitals and Google search results | Google Search Central | Documentation | Google for Developers
https://developers.google.com/search/docs/appearance/core-web-vitals

[15] [16] [17] [18] [20] How to Write Meta Descriptions | Google Search Central | Documentation | Google for Developers
https://developers.google.com/search/docs/appearance/snippet